

Fatma Chihaoui

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[chfatma.github.io/portfolio](https://github.com/chfatma)

Graduate software engineer specializing in Data Science, Artificial Intelligence, and web/mobile development. Top of class for the first three years of my degree, I have gained solid technical experience through hands-on projects in AI (CNN, TensorFlow, XGBoost, LLM), Full-Stack (Spring Boot, React.js, Angular, FastAPI), and cross-platform mobile (Flutter/Dart). Diligent, versatile, and self-driven, I am seeking an engineering position to contribute to high-impact projects.

Technical Skills:

- **Frameworks:** React.js, Angular, Spring Boot, FastAPI, Flask, Django, Flutter.
- **Languages:** Python, Java, TypeScript, Dart, C/C++, SQL, HTML/CSS.
- **Databases:** MySQL, PostgreSQL, SQLite, Redis.
- **Tools:** Git, GitHub/GitLab, Docker, Jupyter Notebook.
- **AI / Data Science:** TensorFlow, Keras, Scikit-learn, XGBoost, Pandas, NumPy, Matplotlib.

Professional Experience

- **StartNow** Tunis
Fullstack Developer – Web & Mobile Real Estate Management Platform (BMS) Apr. 2026 – Present
Development of a full-featured multi-role Building Management System (Super Admin, Admin, Owner, Tenant, Employee).
 - Multi-module React.js frontend: buildings, floors, units, contracts, rents, invoices, complaints, maintenance, leave, payroll, visitors.
 - JWT authentication (in-memory access token, HttpOnly cookie refresh token) and API-driven dynamic RBAC; Axios interceptors for auto-refresh.
 - Real-time messaging module (inter-user chat) via WebSocket (Socket.io) with push notifications; data import module (Excel/CSV); conversational AI Chatbot.
 - Internationalization in 6 languages (EN, FR, AR, DE, PT, ES) with RTL support; Stripe/PayPal payments; visitor OCR; PDF generation.
 - Multi-role Flutter mobile application with real-time synchronization.**Technologies:** React.js, Redux Toolkit, Chakra UI, React Query, Formik, AG Grid, Flutter/Dart, Spring Boot, Socket.io, Stripe, MySQL
- **Final Year Engineering Internship** 4 months
B2B Lead Management & Qualification Platform 2026
Multi-sector B2B leads marketplace: collection, enrichment, ML scoring, and conversational AI agents.
 - Multi-source scraping (Selenium, BeautifulSoup), data.gouv.fr enrichment (SIRENE, NAF, geocoding), and sentiment analysis via local LLM (Ollama).
 - XGBoost v4 scoring module (17 features, Bayesian smoothing, monotonic constraints, v1→v4 versioning) and ML recommendation system (SGDClassifier, online learning, Haversine distance).
 - Multi-role Flutter mobile app (Admin, Buyer, Supplier, Agent), OpenAI GPT-4o (email generation, OCR, business agents), and Stripe.**Technologies:** Python, FastAPI, Flutter/Dart, PostgreSQL, Redis, XGBoost, Scikit-learn, OpenAI API, Stripe, Selenium, BeautifulSoup
- **StartNow** 1 month
Summer Internship – Student Family Management Module 2025
Fullstack web application for managing student families in collaboration with the school administration.
 - Family profile management, parent-teacher meeting scheduling/tracking with automatic attendance rate calculation.
 - Scoring algorithm to determine families eligible for tuition fee discounts.**Technologies:** Spring Boot, React.js, MySQL, JavaScript, HTML/CSS
- **CompassIT** Hammam Sousse
Summer Internship – Parcel Management Web Application 2025
Multi-company web application with delivery driver, customer, and license tracking management.
 - Parcel, customer, and delivery driver management with QR code generation.
 - Parcel location tracking and per-company usage control based on subscribed licenses.**Technologies:** Node.js, React.js, MySQL, JavaScript, TypeScript, HTML/CSS
- **Academic Project**
Real-Time Sign Language Recognition Feb.–Apr. 2024
Complete gesture recognition system combining IoT, AI, and web technologies.
 - IoT: Raspberry Pi and camera to capture hand gestures in real time.
 - AI: Classification model (TensorFlow/Keras) to recognize letters, numbers, and sign language words.
 - Web: Flask interface displaying results with integrated text-to-speech.**Technologies:** Python, TensorFlow, Keras, Flask, Raspberry Pi
- **Academic Project**
Multiplayer Hangman with Binary Search Tree Jan.–Feb. 2024

Multiplayer Hangman game with advanced data structures and network architecture.

- Word storage in a Binary Search Tree (BST); three difficulty levels.
- Multithreaded multiplayer server (C and Python); console interface and graphical/web interface with score database.

Technologies: C, Python, Flask, SQLite, Linux, VMware

CompassIT

Hammam Sousse

2023

Final Year Internship – Olive Oil Classification – Web & Mobile App

Fullstack web and mobile application to classify and trace olive oil test results.

- Oil type classification via specific tests; distinction between PRO CLIENT / PAR CLIENT with dedicated profiles.
- Results saved to MySQL database and unique QR code generated per result for traceability.

Technologies: FastAPI, Angular, Flutter, MySQL, Python, Dart, TypeScript

ItGateTraining

Hammam Sousse

2022

Summer Internship – Data Analyst – COVID-19 Radiological Image Classification

Complete ingestion and classification pipeline for medical images using deep learning, with a dedicated mobile application.

- Dataset preparation and augmentation; CNN model training (VGG) with TensorFlow; SQLite storage and Flask visualization.
- Flutter/Dart mobile application to submit images and view results in real time.

Technologies: Python, TensorFlow, Flask, SQLite, Flutter/Dart, HTML/CSS

Personal Project

Personal Finance Tracker

Mar.–Jun. 2022

Web application for managing and analyzing daily retail transactions.

- Test data generation, SQL ingestion/transformation, transaction analysis, and expense forecasting module.
- Angular (frontend) and Django (backend) web interface for data visualization and alert triggering.

Technologies: Python, TypeScript, SQL, Angular, Django, HTML/CSS

Personal Project

Restaurant Management Mobile Application

Apr.–May 2022

Design and development of a mobile application for managing reservations, payments, and deliveries.

Technologies: Flutter/Dart, Flask, SQL, Python

Education

Institut Supérieur d'Informatique

Tunis El Manar

Engineering Degree – IDISC (Digital Infrastructures and Communication Services)

2023 – 2026

ISMAI Kairouan

Bachelor's Degree in Computer Systems Engineering – Top of Class

2020 – 2023

Certifications

- DevNet Associate – Cisco Networking Academy (2025) [View badge](#)
- CCNA 1, 2, 3 – Cisco Networking Academy (2025) [View badges](#)
- Python for Data Science – DataCamp / CDOSS (2025) [View badge](#)
- Certified Data Science Training – ItgateTraining (2022)

Languages

- **Arabic:** Native **French:** Fluent – Full Professional Proficiency **English:** Fluent – Full Professional Proficiency